

Marko Mladenovic

marko@bugdigger.com | bugdigger.com | github.com/bbugdigger.com

Work Experience

Anti Cheat Consultant	Shindiri Studios – Nis, Serbia	10/2024 – 10/2024
- Tested security of game clients, servers, mobile and web apps, cloud infrastructure and 3rd party integrations - Coordinated with the company and colleagues to decide the overall direction of game integrity - Reverse engineered active threats in the current game cheating/hacking community		
Cheat Developer	GameVandal – Remote	06/2020 – 10/2024
- Designed kernel-mode drivers and userland components (C/C++, x86/x64 assembly) on Windows platform - Reverse engineered commercial anti cheat systems (Battleeye, EasyAntiCheat) to understand weaknesses - Worked on whole cheat deployment workflow		
Reverse Engineer Internship	AST – Nis, Serbia	05/2024 – 08/2024
- Reverse engineered and found vulnerabilities on a IP camera's firmware - Investigated malicious PDFs and DOCX files and reversed engineered the JavaScript and PowerShell obfuscations		
Java Backend Developer	FI3xx GmbH – Vienna, Austria	05/2022 – 07/2023
- Backend developer (Tech stack: Java, Spring, Hibernate, JPA, Elasticsearch, Postgres, Redis) - Maintained and expanded existing code base with occasional new feature development - Found SSRF through pdf generation security vulnerability and remediated it - Performed secure code reviews using SonarQube; identified and mitigated XXE and SQL injection vulnerabilities		

Certifications & Achievements

- Univerzitet Metropolitani, Software Engineering**
- Zero2Automated** - courses.offset.net/adv-malware-analysis-course **In Progress**
 - Focused on advanced static and dynamic analysis of real-world malware, unpacking and deobfuscation techniques, tracking malware families, signature creation, and automating malware analysis workflows
- Ret2 Systems Binary Exploitation Course** - wargames.ret2.systems/course **February 2024**
 - Reverse engineered an algorithm that generates a software license
 - Found race condition vulnerability in concurrent garbage collector. Challenge was inspired by the JavaScriptCore vulnerability presented at Pwn2Own 2018 competition
- ZDE VR Course** - zerodayengineering.com/training/universal-vulnerability-research.html **February 2024**
 - Got essential knowledge and skills required to conduct modern low-level security research, vulnerability exploit development, secure software engineering and hardening

Skills

- Programming languages (Java, C/C++, Rust, JavaScript, Python, x86-64 and ARM64 assembly)
- Development Tools, Practices and Environments: Docker, Git, Jenkins, Agile/Scrum, CI/CD, Linux, Windows
- Web exploitation techniques (XSS, SQL injections, SSRF, Directory traversal, XXE, ...)
- Reverse engineering, static analysis (ghidra, binary ninja, ida), and dynamic analysis (x64dbg, windbg, gdb)
- Ability to bypass common exploitation techniques (ASLR, DEP, PIE, Stack cookies)
- Systems programming in context of writing compilers and kernel drivers (Windows, Linux)
- Malware analysis (unpacking, API resolving, emulation, symbolic execution, deobfuscation, devirtualization)
- Familiarity with reversing obfuscation, encryption and anti-tamper technologies
- Ability to quickly tackle and get familiar with unfamiliar and large codebases

Projects

- Rust ARM64 kernel for Raspberry Pi 3** - <https://github.com/bbugdigger/armRustOS> **In Progress**
 - Developing ARM64 kernel in rust for Raspberry Pi 3